

Notes: Chari, Kehoe, and McGrattan (Econometrica, 2007)

Business Cycle Accounting

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1 One-sentence summary

Business cycle accounting decomposes observed fluctuations into efficiency, labor, investment, and government-consumption wedges, and finds that U.S. downturns are mainly accounted for by efficiency and labor wedges, while investment wedges play only a tertiary role.

2 Big picture

2.1 Main question

Which classes of business-cycle models are promising for explaining large aggregate fluctuations? The paper proposes a diagnostic method: translate many detailed frictions into wedges in a prototype growth model, measure those wedges in the data, and ask which wedges account for observed movements in output, labor, and investment.

2.2 Core method

The method has two pieces.

- **Equivalence results:** many detailed models are observationally equivalent to a prototype growth model with time-varying wedges.
- **Accounting procedure:** measure wedges from data, feed each wedge back into the prototype model, and ask how much of the data each wedge explains.

2.3 Main findings

For the Great Depression and the 1982 recession, efficiency and labor wedges jointly account for essentially all movements in output, labor, and investment. Investment wedges do little or move output in the wrong direction. Government-consumption wedges matter little.

2.4 Why the result matters

The paper does not directly estimate a fully structural Depression model. Instead, it tells model builders where to look. If a candidate model's primary friction maps into an investment wedge, the paper argues that the model is unlikely to be a first-order account of U.S. business cycles. Models mapping into efficiency and labor wedges are more promising.

3 Incremental contribution of the paper

3.1 A diagnostic framework for business-cycle model building

The paper creates a method for sorting broad classes of business-cycle mechanisms before building a fully specified model. This is a methodological contribution: it gives macroeconomists an accounting device for narrowing the model space.

3.2 Equivalence between detailed frictions and prototype wedges

The paper shows that input-financing frictions, sticky wages, investment frictions, and open-economy borrowing shocks can be represented as wedges in a standard growth model. This links micro-founded frictions to measurable macro wedges.

3.3 Evidence against investment-wedge-centered explanations

A central substantive contribution is the finding that investment wedges do not account well for the Great Depression or the 1982 recession. This challenges explanations based mainly on financial frictions that appear primarily as investment distortions.

3.4 Evidence in favor of efficiency and labor wedges

Efficiency and labor wedges together account for the data in both an extreme downturn and a more typical postwar recession. This points toward mechanisms such as input-financing frictions, sticky wages, monopoly/union distortions, and other forces that manifest as productivity or labor wedges.

3.5 Robustness across alternative prototype specifications

The paper also shows that its key conclusions survive changes such as variable capital utilization, labor-supply elasticity changes, and investment adjustment costs. This helps distinguish true frictions from specification differences.

4 Conceptual framework

4.1 Prototype economy

The prototype is a growth model with representative households and firms. Output is produced with capital and labor, but the equilibrium conditions are distorted by four wedges. These wedges look like productivity shocks, labor taxes, investment taxes, and government consumption.

4.2 Four wedges

- **Efficiency wedge:** works like a time-varying productivity term, denoted A_t .
- **Labor wedge:** works like a time-varying labor income tax, denoted $1 - \tau_{lt}$.
- **Investment wedge:** works like a time-varying tax on investment, often written $1/(1 + \tau_{xt})$.
- **Government-consumption wedge:** works like exogenous absorption that moves resources away from private use.

4.3 Interpreting wedges

The wedges are not literal taxes in most applications. They are residual distortions in equilibrium conditions. A sticky-wage model may show up as a labor wedge; some financial frictions may show up as an efficiency wedge; credit frictions of the Bernanke-Gertler-Gilchrist type can show up as an investment wedge.

5 Model setup and wedge measurement

5.1 Resource constraint

A stylized prototype resource constraint is

$$C_t + X_t + G_t = A_t F(K_t, L_t).$$

The measured efficiency wedge is the residual object that makes this equation and the production function match observed output, capital, and labor.

5.2 Labor first-order condition

The labor wedge appears in the household's intratemporal condition:

$$-\frac{U_{L,t}}{U_{C,t}} = (1 - \tau_{lt})A_t F_L(K_t, L_t).$$

If measured labor is low relative to consumption and productivity, the labor wedge deteriorates.

5.3 Investment Euler equation

The investment wedge appears in the intertemporal condition:

$$U_{C,t}(1 + \tau_{xt}) = \beta E_t [U_{C,t+1} \{A_{t+1} F_K(K_{t+1}, L_{t+1}) + 1 - \delta\}].$$

A worsening investment wedge makes investment expensive relative to future returns.

5.4 State-space measurement

The paper treats wedges as stochastic processes and estimates vector autoregressions for measured wedges. The estimated processes are then used to run accounting experiments in which wedges are included one at a time or removed one at a time.

6 Accounting procedure

6.1 Step 1: measure wedges

The authors use the prototype model equilibrium conditions and U.S. data to infer the time series of wedges. By construction, all measured wedges together exactly reproduce the observed data in the prototype economy.

6.2 Step 2: feed wedges back into the model

The authors then run counterfactual model simulations:

- include only the efficiency wedge;
- include only the labor wedge;
- include only the investment wedge;
- include all wedges except one wedge;
- compare predicted output, labor, and investment with the data.

6.3 Step 3: assign accounting importance

A wedge is important if a model with that wedge reproduces the movements in the data, or if removing that wedge destroys the model's ability to match the data. This is an accounting decomposition, not a causal experiment.

7 Main empirical applications

7.1 Great Depression, 1929-1939

The Great Depression is the paper's most demanding test. Output, labor, and investment collapse between 1929 and 1933 and then recover slowly. The efficiency and labor wedges together match this pattern well. The investment wedge moves output in the wrong direction.

7.2 1982 recession

The 1982 recession provides a more typical postwar downturn. The same qualitative conclusion emerges: efficiency and labor wedges matter, investment wedges do little, and government consumption plays little role.

7.3 Entire postwar period

For 1959-2004, summary statistics show that the investment wedge matters somewhat more than in the 1982 episode, but still much less than efficiency and labor wedges.

7.4 Alternative specifications

The authors explore variable capital utilization and investment adjustment costs. These robustness exercises do not overturn the main conclusion: investment-wedge explanations remain weak relative to efficiency- and labor-wedge explanations.

8 Identification and interpretation

8.1 What the paper identifies

Business cycle accounting identifies which wedges in the prototype economy can reproduce observed fluctuations. It identifies a mapping from data to wedges and from wedges to model-generated fluctuations.

8.2 What the paper does not identify

The paper does not directly identify the true primitive friction. Many detailed models can map into the same wedge. For example, an efficiency wedge could represent technology, input-financing distortions, capacity utilization mismeasurement, or other forces.

8.3 Observable implications

The main observable implication is wedge priority. If the investment wedge cannot account for output, labor, and investment dynamics, then models whose only important friction maps into investment wedges are unlikely to be quantitatively successful.

8.4 Main limitation

Wedges are diagnostic objects, not structural shocks. The method is best understood as a way of disciplining future model building rather than as a final explanation of business cycles.

9 Tables and figures

9.1 Table I: Parameters of the Vector AR(1) Stochastic Process in Two Historical Episodes

Read: Table I reports the estimated stochastic processes for the measured wedges in annual data for 1901-1940 and quarterly data for 1959:1-2004:3. The matrices govern the dynamic interactions among wedges and are used to simulate the prototype model.

Economic message: The table supplies the time-series machinery behind the accounting experiments. The model needs a stochastic law of motion for wedges so that agents form expectations.

Detailed interpretation: The table is technical but important. It shows that the accounting exercise is not a purely static decomposition. Wedges are persistent and correlated, and those dynamics affect predicted consumption, investment, labor, and output.

TABLE I
PARAMETERS OF THE VECTOR AR(1) STOCHASTIC PROCESS IN TWO HISTORICAL EPISODES^a
(ESTIMATED USING MAXIMUM LIKELIHOOD WITH U.S. DATA^b)

Coefficient Matrix P on Lagged States					Coefficient Matrix Q , Where $V = QQ'$				
A. Annual Data, 1901-1940									
.732	-.0521	-.317	0		.0575	0	0	0	
(.470, .856)	(-.0364, .142)	(-.716, .130)			(.0440, .0666)				
-.150	.104	.390	0		-.00561	.0555	0	0	
(-.339, .0504)	(.908, 1.10)	(-.0751, .782)			(-.0216, .00952)	(.0378, .0643)			
-.0114	-.0197	.0731	0		.000299	-.000253	.0369	0	
(-.384, .260)	(-.262, .126)	(-.363, .296)			(-.0308, .0230)	(-.0167, .0121)	(.0194, .0489)		
0	0	0	.750		0	0	0	.221	
			(.424, .814)					(.145, .276)	
Means of states = [.541 (.503, .591), -.190 (-.271, -.0867), .286 (.216, .364), -2.79 (-2.95, -2.55)]									
B. Quarterly Data, 1959:1-2004:3									
.980	-.0138	-.0117	.0192		.0116	0	0	0	
(.944, .984)	(-.0192, .00222)	(-.0129, -.00605)	(.0125, .0259)		(.0105, .0126)				
-.0330	.956	-.0451	.0569		.00141	.00644	0	0	
(-.0396, -.0061)	(.920, .959)	(-.0512, -.0286)	(.0473, .0677)		(.000462, .00232)	(.00567, .00695)			
-.0702	-.0460	.896	.104		-.0105	.00103	.0158	0	
(-.1087, -.0672)	(-.0612, -.0304)	(.879, .907)	(.0817, .112)		(-.0141, -.00779)	(-.00278, .00266)	(.0133, .0190)		
.00481	-.00811	.0488	.971		-.000575	.00611	.0142	.00458	
(-.0278, .0116)	(-.0158, .0157)	(.0371, .0643)	(.954, .974)		(-.00219, .00132)	(.00383, .00760)	(.0121, .0154)	(.00386, .00554)	
Means of states = [-.0239 (-.0301, -.0137), .328 (.322, .336), .483 (.473, .495), -1.53 (-1.55, -1.52)]									

^aTo ensure stationarity, we add to the likelihood function a penalty term proportional to $\max(\lambda_{\max}) - .995, 0^2$, where $\lambda_{\max}$ is the maximal eigenvalue of P . Numbers in parentheses are 90% confidence intervals for a bootstrapped distribution with 500 replications. To ensure that the variance-covariance matrix V is positive semidefinite, we estimate Q rather than $V = QQ'$.

^bFor the sources of basic data, see Chari, Kehoe, and McGrattan (2006).

Table I: Parameters of the Vector AR(1) stochastic process in two historical episodes. The table reports coefficient and innovation matrices used to generate expectations over future wedges.

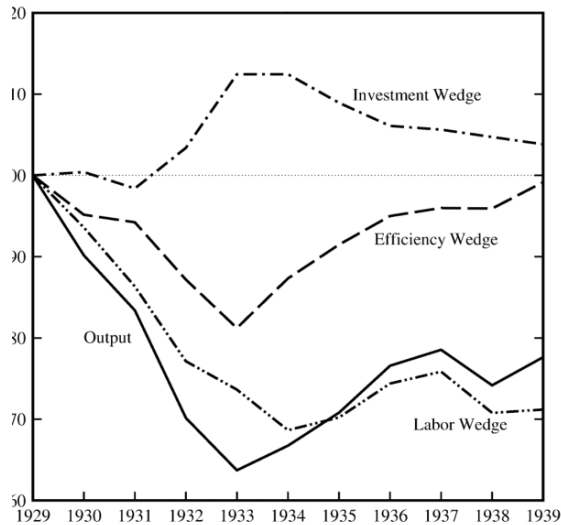
9.2 Figure 1: U.S. output and measured wedges; Figure 2: Great Depression one-wedge predictions

Read: Figure 1 shows Great Depression output and the three main measured wedges. Figure 2 shows model predictions when only the efficiency wedge or only the labor wedge is active.

Economic message: Efficiency and labor wedges deteriorate strongly during the Depression and both generate substantial output declines when fed into the model.

Detailed interpretation: Figure 1 identifies the candidate distortions. Figure 2 evaluates two of them. The efficiency wedge accounts for much of the 1929-1933 output drop, while the labor wedge helps account for the depth and persistence of the downturn.

one wedge. In terms of the data, note that labor decline



U.S. output and three measured wedges (annually; normalized to eq
(a) Figure 1: U.S. output and three measured wedges, Great Depression.

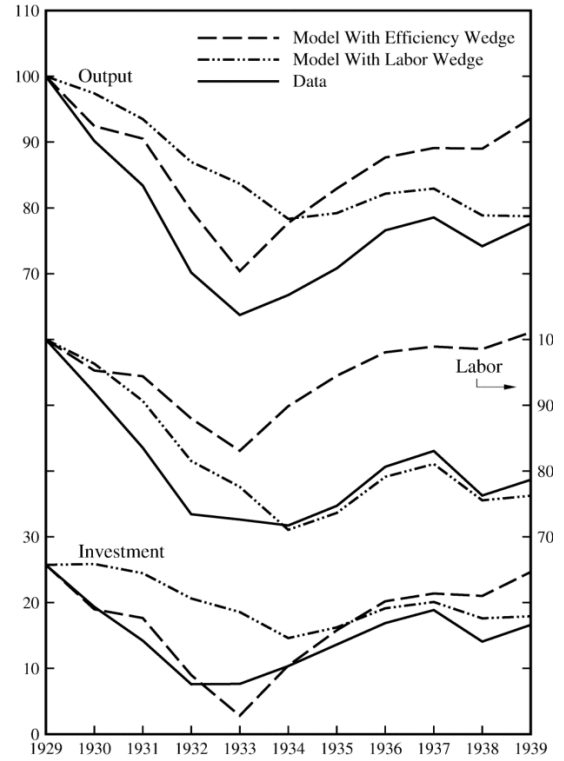


FIGURE 2.—Data and predictions of the models with just one wedge

1929 to 1933 and stays relatively low for the rest of the dec
(b) Figure 2: Data and predictions with just one wedge.

9.3 Figure 3: Great Depression investment wedge; Figure 4: Great Depression all-wedges-but-one predictions

Read: Figure 3 feeds in only the investment wedge. Figure 4 asks what happens when one wedge is removed while the others remain.

Economic message: The investment wedge performs poorly. It predicts output movements that do not match the data and produces the consumption anomaly discussed in the paper.

Detailed interpretation: These figures are the main evidence against investment-wedge-centered explanations of the Great Depression. Removing efficiency or labor wedges matters; relying on the investment wedge alone does not.

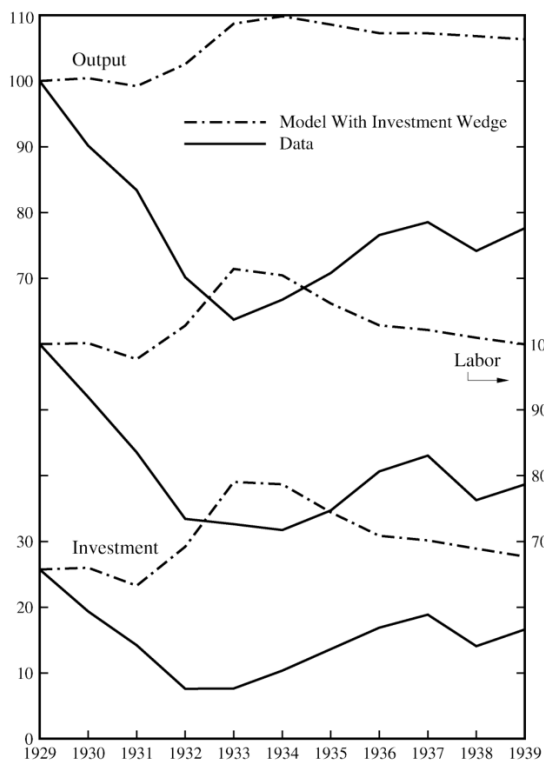


FIGURE 3.—Data and predictions of the model with just the investment w

this experiment, we choose the investment wedge to be as large

(a) Figure 3: Great Depression predictions with just the investment wedge.

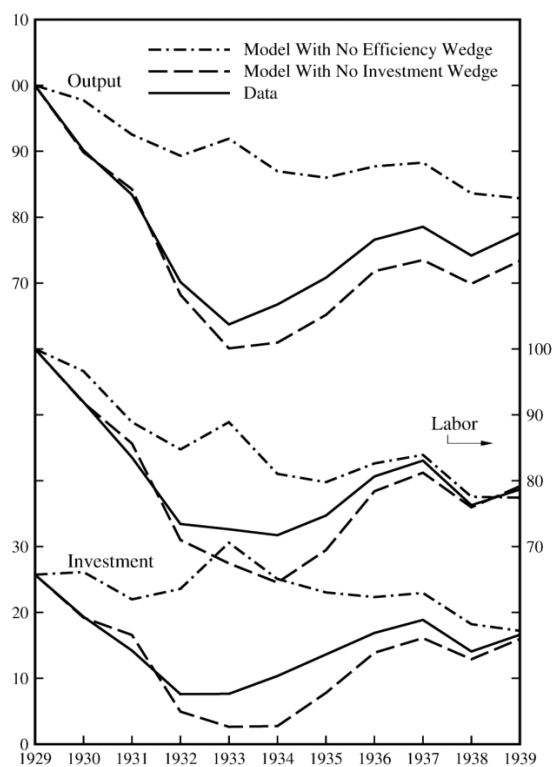


FIGURE 4.—Data and predictions of the models with all wedges but one.

(b) Figure 4: Great Depression predictions with all wedges but one.

9.4 Figure 5: 1982 measured wedges; Figure 6: 1982 one-wedge predictions

Read: Figure 5 plots output and wedges for 1979:1-1985:4. Figure 6 repeats the one-wedge accounting exercise for the 1982 recession.

Economic message: The same broad pattern appears outside the Great Depression: efficiency and labor wedges matter more than investment wedges.

Detailed interpretation: The 1982 recession strengthens external validity. The conclusions are not special to the Depression's extreme environment.

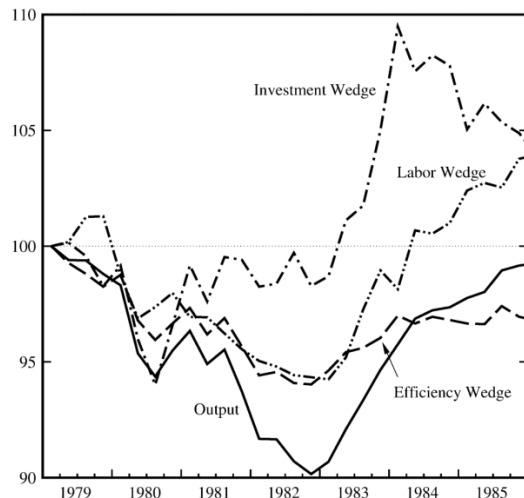


FIGURE 5.—U.S. output and three measured wedges (quarterly, 1979:1–1985:4; 100 in 1979:1).

between 1979 and 1982, and by 1985 is still a little more than 3% below 1979. The labor wedge also worsens from 1979 to 1982, but it improves somewhat by 1985. The investment wedge meanwhile fluctuates until 1982, then rises sharply.

(a) Figure 5: Output and measured wedges around the 1982 recession.

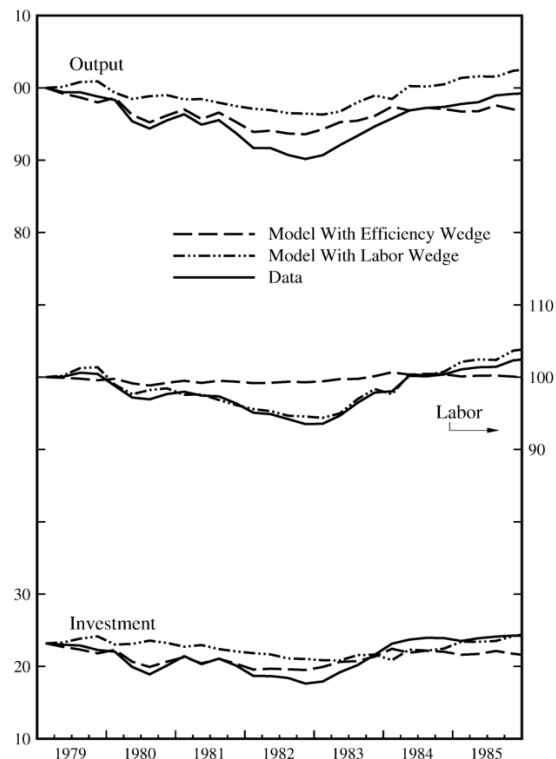


FIGURE 6.—Data and predictions of the models with just one wedge.

(b) Figure 6: 1982 predictions with just one wedge.

9.5 Figure 7: 1982 investment wedge; Figure 8: 1982 all-wedges-but-one predictions

Read: Figure 7 isolates the investment wedge for the 1982 recession. Figure 8 removes one wedge at a time.

Economic message: The investment wedge generates essentially no useful business-cycle movement in this episode. The efficiency and labor wedges remain the key drivers.

Detailed interpretation: Together with Figures 5-6, these figures show that the paper's model-selection advice extends to ordinary recessions as well as the Great Depression.

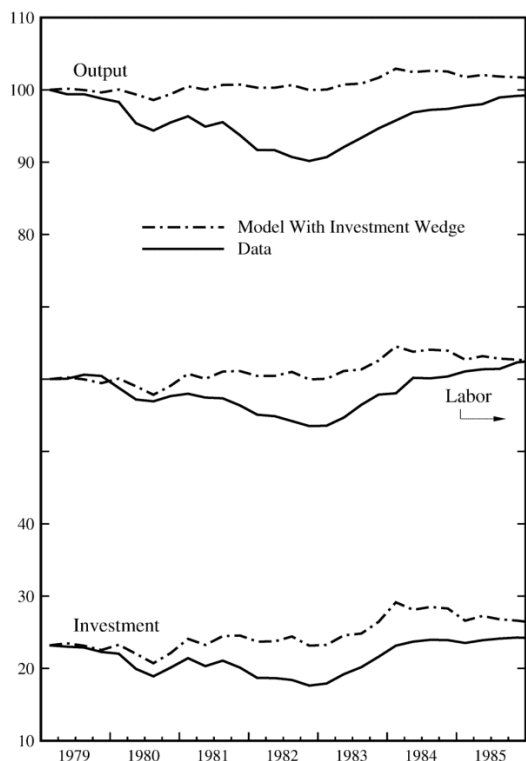


FIGURE 7.—Data and predictions of the model with just the investment w

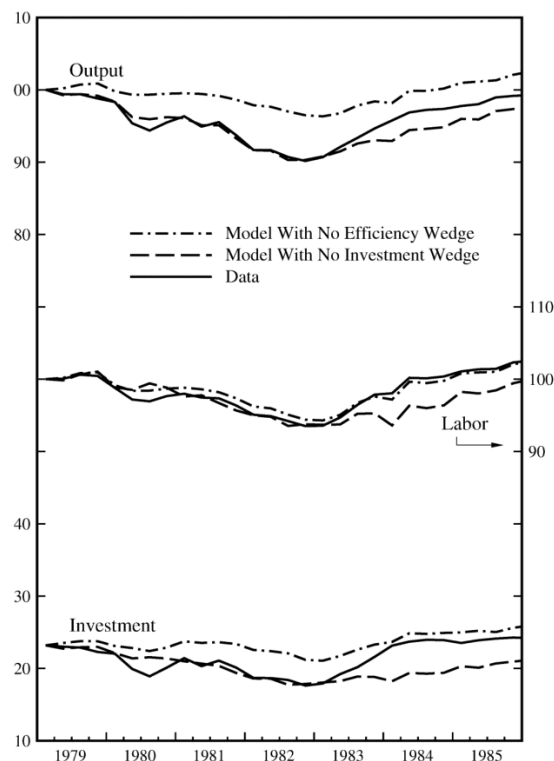


FIGURE 8.—Data and predictions of the models with all wedges but one.

(a) Figure 7: 1982 predictions with just the investment wedge.

(b) Figure 8: 1982 predictions with all wedges but one.

9.6 Table II: Postwar statistics; Table III: Postwar output movements due to wedges

Read: Tables II and III summarize HP-filtered moments over 1959-2004. They report standard deviations and correlations of wedge-induced output movements with the data.

Economic message: The postwar analysis compresses many cycles into summary statistics. It shows that investment wedges matter somewhat more over the full postwar sample than in 1982, but still less than efficiency or labor wedges.

Detailed interpretation: The tables are useful because visual accounting is episode-specific. These statistics show that the paper's ranking of wedges is not driven by only two episodes.

TABLE II
PROPERTIES OF THE WEDGES, 1959:1–2004:3^a

A. Summary Statistics						
Wedges	Standard Deviation Relative to Output	Cross Correlation of Wedge with Output at Lag $k =$				
		-2	-1	0	1	2
Efficiency	.63	.65	.76	.85	.60	.35
Labor	.92	.52	.65	.71	.73	.68
Investment	1.18	.44	.48	.47	.30	.09
Government consumption	1.51	-.42	-.42	-.33	-.24	-.11
B. Cross Correlations						
Wedges (X, Y)		Cross Correlation of X with Y at Lag $k =$				
		-2	-1	0	1	2
Efficiency, labor		.57	.48	.30	.28	.16
Efficiency, investment		.31	.46	.61	.47	.35
Efficiency, government consumption		-.27	-.33	-.34	-.35	-.31
Labor, investment		-.07	.11	.18	.37	.46
Labor, government consumption		-.02	-.22	-.38	-.47	-.50
Investment, government consumption		-.60	-.73	-.88	-.70	-.51

^aSeries are first logged and detrended using the HP filter.

output, both contemporaneously and for several leads and lags. In contrast, the government consumption wedge is somewhat negatively correlated with output, both contemporaneously and for several leads and lags. (Note that the government consumption wedge is the sum of government consumption and net exports, and that net exports are negatively correlated with output.) Panel B of Table II shows that the cross correlations of the efficiency, labor, and investment wedges are generally positive.

Table III summarizes various statistics of the movements of output over this period due to each wedge. Consider panel A and focus first on the output fluctuations due to the efficiency wedge. Table III shows that output movements due to this wedge have a standard deviation that is 73% of that of output in the data. These movements are highly positively correlated with output in the data, both contemporaneously and for several leads and lags. These statistics

(a) Table II: Standard deviations and correlations, postwar period.

TABLE III
PROPERTIES OF THE OUTPUT COMPONENTS, 1959:1–2004:3^a

A. Summary Statistics						
Output Components	Standard Deviation Relative to Output	Cross Correlation of Wedge with Output at Lag $k =$				
		-2	-1	0	1	2
Efficiency	.73	.65	.75	.83	.57	.31
Labor	.59	.44	.59	.68	.74	.74
Investment	.31	.33	.37	.40	.25	.07
Government consumption	.40	-.45	-.45	-.39	-.25	-.08
B. Cross Correlations						
Output Components (X, Y)		Cross Correlation of X with Y at Lag $k =$				
		-2	-1	0	1	2
Efficiency, labor		.54	.41	.18	.15	.04
Efficiency, investment		.30	.44	.60	.40	.28
Efficiency, government consumption		-.34	-.45	-.56	-.48	-.39
Labor, investment		-.17	-.03	-.03	.20	.29
Labor, government consumption		.14	-.03	-.13	-.31	-.40
Investment, government consumption		-.49	-.63	-.87	-.66	-.48

^aSeries are first logged and detrended using the HP filter.

tively correlated with it. Output due to the investment wedge alone fluctuates less than one-third as much as output in the data and is somewhat positively correlated with it. Finally, output due to the government consumption wedge alone fluctuates about 40% as much as output in the data and is somewhat negatively correlated with it. In panel B of Table III, we see that output movements due to the efficiency and labor wedges as well as the efficiency and investment wedges are positively correlated, and that the cross correlations of output movements due to the other wedges are mostly essentially zero or negative.

All of our analyses using business cycle accounting thus seem to lead to the same conclusion: to study business cycles, the most promising detailed models to explore are those in which frictions manifest themselves primarily as effi-

(b) Table III: Output movements due to wedges, postwar period.

9.7 Figure 9: Variable utilization measured wedges; Figure 10: Variable utilization one-wedge predictions

Read: Figure 9 remeasures wedges under variable capital utilization. Figure 10 runs the one-wedge prediction experiment under that specification.

Economic message: Allowing utilization to vary changes the measured efficiency wedge but does not rescue the investment wedge explanation.

Detailed interpretation: The exercise separates genuine frictions from specification differences. Some of what the benchmark calls an efficiency wedge may reflect utilization, but the main diagnostic conclusion remains intact.

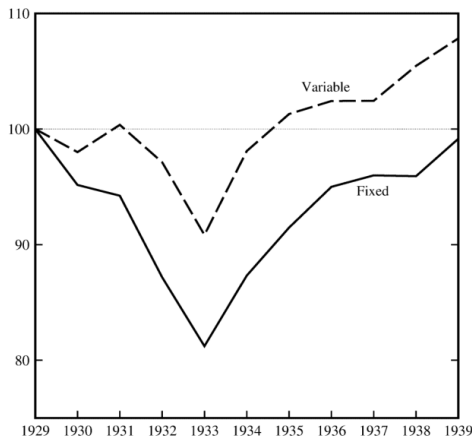


FIGURE 9.—Varying the capital utilization specification during the Great Depression period: efficiency wedges for two capital utilization specifications.

Comparing this to Figure 3, we see that with variable capital utilization, the investment wedge drives output the wrong way to an even greater extent than in the fixed case. (a) Figure 9: Wedges under variable capital utilization.

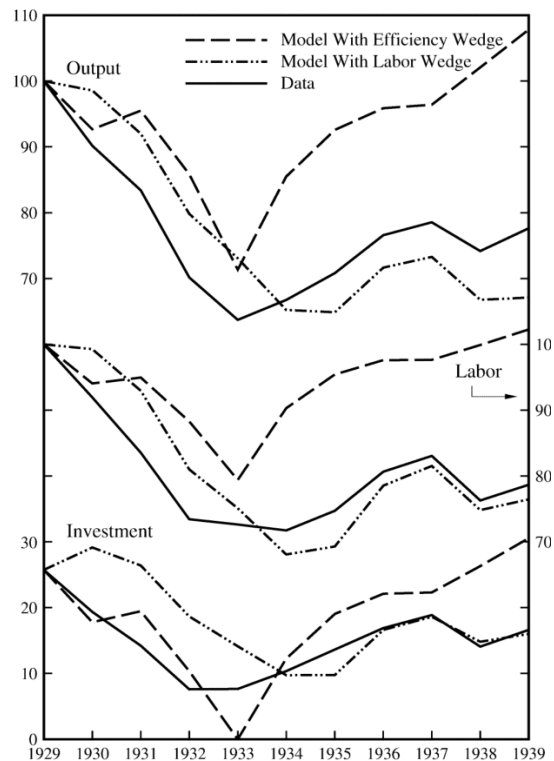


FIGURE 10.—Data and predictions of the models with variable capital utilization: efficiency wedges.

(b) Figure 10: Variable-utilization predictions with just one wedge.

9.8 Figure 11: Variable utilization investment wedge; Figure 12: Fixed versus variable utilization

Read: Figure 11 isolates the investment wedge with variable utilization. Figure 12 compares fixed and variable utilization when efficiency and labor wedges are combined.

Economic message: Variable utilization does not make the investment wedge important. The combined efficiency-labor explanation remains central.

Detailed interpretation: The figures show that one cannot overturn the paper's conclusion by changing the utilization specification. If anything, the investment wedge becomes even less convincing.

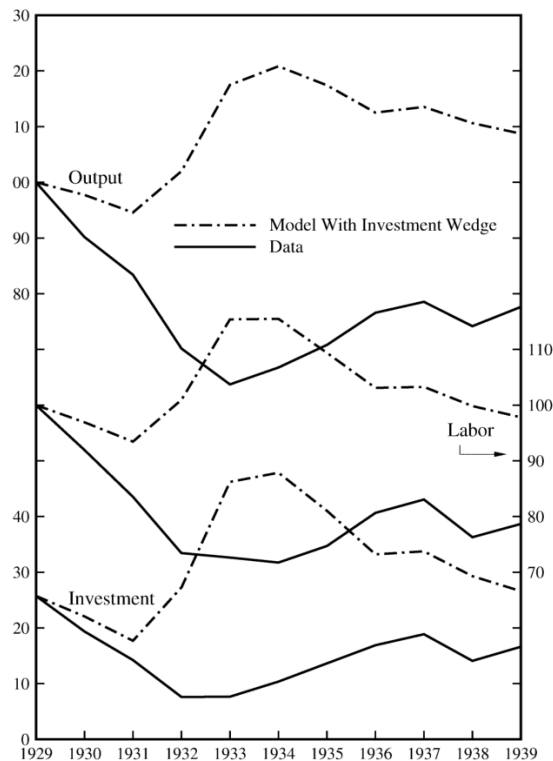


Figure 11.—Data and predictions of the model with variable capital utilization and investment wedge.

(a) Figure 11: Variable-utilization predictions with the investment wedge.

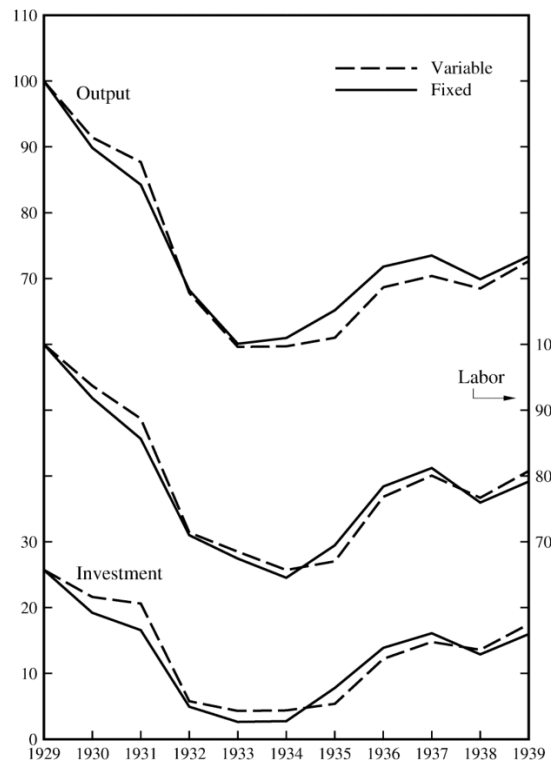


Figure 12.—Predictions of the models with fixed and variable capital utilization and investment wedge.

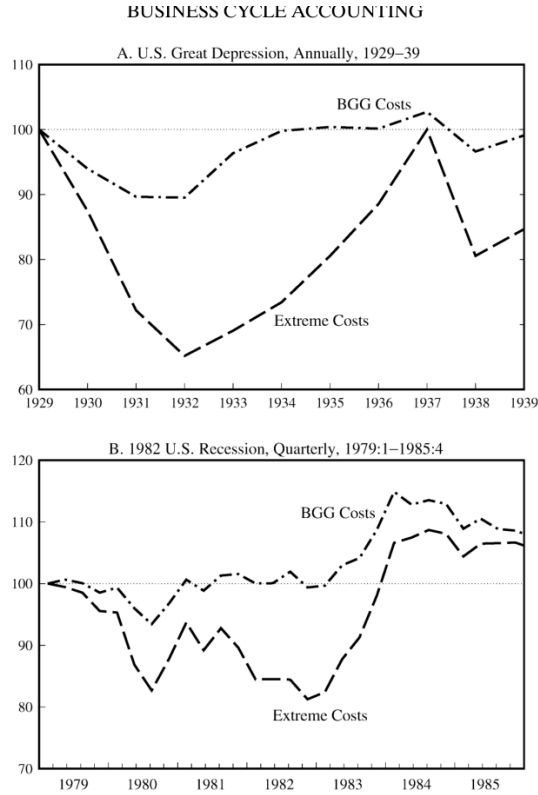
(b) Figure 12: Fixed versus variable utilization with efficiency and labor wedges.

9.9 Figure 13: Adjustment-cost investment wedges; Figure 14: Adjustment-cost output predictions

Read: Figure 13 measures investment wedges under two adjustment-cost specifications. Figure 14 shows the corresponding output predictions.

Economic message: Investment adjustment costs can make the investment wedge look more plausible in some dimensions, but the investment wedge still does not become the primary account of U.S. downturns.

Detailed interpretation: The final robustness exercise is aimed at financial-friction models that operate through costly investment adjustment. The paper allows this channel but still concludes that efficiency and labor wedges are the more promising focus.



(a) Figure 13: Measured investment wedges under adjustment-cost specifications.

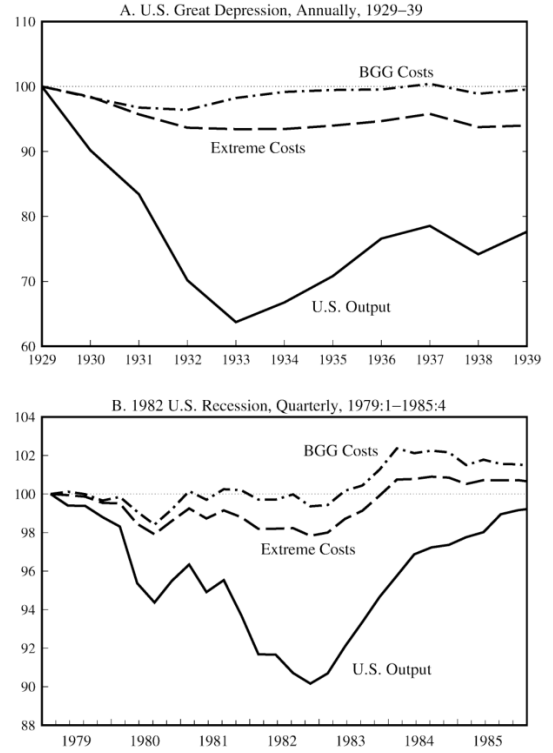


FIGURE 14.—U.S. output and predictions of model with alternative adjustment

(b) Figure 14: Output predictions under adjustment-cost specifications.

10 Short summary

- Business cycle accounting maps many detailed frictions into four wedges in a prototype growth model.
- The four wedges are efficiency, labor, investment, and government-consumption wedges.
- The accounting procedure measures wedges and feeds them back into the model one at a time and in combinations.
- The Great Depression is largely accounted for by efficiency and labor wedges.
- The investment wedge performs poorly for the Great Depression and creates the wrong movement in output in key periods.
- The 1982 recession delivers the same qualitative result.
- Postwar summary statistics show investment wedges play a smaller role than efficiency and labor wedges.
- Robustness checks with utilization and adjustment costs do not overturn the conclusion.
- The paper's main advice is to build models whose primitive frictions manifest as efficiency and labor wedges.

11 Conclusion

11.1 Core conclusion

Chari, Kehoe, and McGrattan provide a diagnostic framework for macroeconomic model building. They show that U.S. business cycles are best accounted for by efficiency and labor wedges, while investment wedges play a much smaller role.

11.2 Economic intuition

The method asks what must be distorted in a standard growth model to reproduce the data. If a wedge does not reproduce the key movements in output, labor, and investment, then models that rely primarily on that wedge are unlikely to explain the cycle.

11.3 Implication for financial-friction models

The paper does not say financial frictions are unimportant. It says that financial frictions that mainly appear as investment wedges are unlikely to be the primary account of U.S. recessions. Financial frictions that appear as efficiency wedges may be much more promising.

11.4 Limitations

Wedges are reduced-form diagnostic objects. They do not uniquely identify primitive shocks or frictions. The method tells researchers where to look, not the final structural model to use.

11.5 Takeaway

Business cycle accounting is a disciplined screening device. It narrows the search for successful business-cycle models by asking which equilibrium distortions are quantitatively large enough to account for the data.